

$$C_{\phi}^{(12)} [{}^{(3)}i1-i2_{-} \rightarrow \frac{1}{16\pi^2} \left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,0} [m_2, \tilde{\mu}] \delta_{i1i2} - \frac{1}{6} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] \delta_{i1i2} + \right. \\ \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} - \frac{1}{6} \tilde{\mu} g_2^2 \overline{y_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] \delta_{i1i2} + \\ \frac{3}{8} \tilde{\mu} g_2^2 \overline{y_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} - \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] \delta_{i1i2} + \\ \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] \delta_{i1i2} - \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_u^{\text{r}}] \delta_{i1i2} - \\ \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} - \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} + \\ \frac{3}{4} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} - \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] \delta_{i1i2} + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} - \\ \frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,1,0} [\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{3} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] \delta_{i1i2} + \\ \frac{7}{8} m_2 \tilde{\mu} g_2^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{2} m_2 \tilde{\mu} g_2^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} - \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_e^{\text{p}}, m_l^{\text{i2}}] - \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, m_e^{\text{p}}, m_l^{\text{i2}}] + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_e^{\text{p}}, m_l^{\text{i2}}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,0} [m_1, m_e^{\text{p}}, \tilde{\mu}] - \\ \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{3,1,1,-1} [m_1, m_e^{\text{p}}, \tilde{\mu}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, m_l^{\text{i2}}, m_e^{\text{p}}] - \\ \frac{3}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{\text{p}}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_l^{\text{i2}}] + \\ \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_e^{\text{p}}] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^{\text{i1}}] \delta_{i1i2} - \\ \frac{3}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^{\text{i2}}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_l^{\text{i2}}] + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_l^{\text{i2}}] + \frac{3}{8} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_2, m_l^{\text{i2}}] - \\ \frac{3}{8} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_2, m_l^{\text{i2}}] + \frac{1}{16} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{1,1,1,1,0} [m_1, m_e^{\text{p}}, m_l^{\text{i1}}, \tilde{\mu}] - \\ \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_1, m_e^{\text{p}}, m_l^{\text{i1}}, \tilde{\mu}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \\ \text{LF}_{1,1,1,1,0} [m_1, m_e^{\text{p}}, m_l^{\text{i2}}, \tilde{\mu}] + \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_1, m_e^{\text{p}}, m_l^{\text{i2}}, \tilde{\mu}] + \\ \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{1,1,1,1,0} [m_2, m_e^{\text{p}}, m_l^{\text{i1}}, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \\ \text{LF}_{2,1,1,1,-1} [m_2, m_e^{\text{p}}, m_l^{\text{i1}}, \tilde{\mu}] + \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{1,1,1,1,0} [m_2, m_e^{\text{p}}, m_l^{\text{i2}}, \tilde{\mu}] + \\ \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_2, m_e^{\text{p}}, m_l^{\text{i2}}, \tilde{\mu}] - \frac{1}{16} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \\ \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_1, m_l^{\text{i1}}, m_l^{\text{i2}}] + \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_1, m_l^{\text{i1}}, \tilde{\mu}] - \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_1, m_l^{\text{i2}}, \tilde{\mu}] + \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \\ \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_2, m_l^{\text{i1}}, m_l^{\text{i2}}] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_2, m_l^{\text{i1}}, \tilde{\mu}] - \\ \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_e^{\text{p}}, m_2, m_l^{\text{i2}}, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \\ \text{LF}_{2,1,1,1,-1} [m_l^{\text{i1}}, m_1, m_e^{\text{p}}, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_l^{\text{i1}}, m_2, m_e^{\text{p}}, \tilde{\mu}] + \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [m_l^{\text{i2}}, m_1, m_e^{\text{p}}, \tilde{\mu}] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \\ \text{LF}_{2,1,1,1,-1} [m_l^{\text{i2}}, m_2, m_e^{\text{p}}, \tilde{\mu}] + \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} a_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_e^{\text{p}}, m_l^{\text{i1}}] - \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{\text{i2p}}} y_e^{\text{i1p}} \text{LF}_{2,1,1,1,-1} [\tilde{\$$